

## HOMEWORK R

### DATASET: nasdaq\_65

#### ESERCIZIO 1

**Calcolare i rendimenti dei sette indici usando la formula  $(\text{prezzo attuale})/(\text{prezzo precedente})-1$ .**

```
> setwd("C:\\Users\\Jeanette\\Downloads")
> nasdaq = read.csv("nasdaq_65.csv")
> library(dplyr)
> library(lmtest)
> library(car)
> library(olsrr)
> library(tseries)
> library(aTSA)
> nasdaq$rasia=nasdaq$asia/lag(nasdaq$asia)-1
> nasdaq$rema=nasdaq$emea/lag(nasdaq$emea)-1
> nasdaq$reuro=nasdaq$euro/lag(nasdaq$euro)-1
> nasdaq$rglobal=nasdaq$global/lag(nasdaq$global)-1
> nasdaq$rlatin=nasdaq$latin/lag(nasdaq$latin)-1
> nasdaq$ruk=nasdaq$uk/lag(nasdaq$uk)-1
> nasdaq$rus=nasdaq$us/lag(nasdaq$us)-1
```

**a) Riportare media e deviazione standard dei rendimenti nei vari indici. Ci sono indici dominati in media-varianza?**

**→ calcolo il rendimento medio giornaliero e la deviazione standard dell'indice Asia**

```
> mean(nasdaq$rasia, na.rm = TRUE)
[1] 0.001911396
> sd(nasdaq$rasia, na.rm = TRUE)
[1] 0.05519819
```

**→ calcolo il rendimento medio giornaliero e la deviazione standard dell'indice EMEA**

```
> mean(nasdaq$rema, na.rm = TRUE)
[1] 0.002979059
> sd(nasdaq$rema, na.rm = TRUE)
[1] 0.06592001
```

**→ calcolo il rendimento medio giornaliero e la deviazione standard dell'indice Eurozone**

```
> mean(nasdaq$reuro, na.rm = TRUE)
[1] 0.003724008
> sd(nasdaq$reuro, na.rm = TRUE)
[1] 0.07476857
```

**→ calcolo il rendimento medio giornaliero e la deviazione standard dell'indice Global**

```
> mean(nasdaq$rglobal, na.rm = TRUE)
[1] 0.003457024
> sd(nasdaq$rglobal, na.rm = TRUE)
[1] 0.08007527
```

**→ calcolo il rendimento medio giornaliero e la deviazione standard dell'indice Latin America**

```
> mean(nasdaq$rlatin, na.rm = TRUE)
[1] 0.0007770978
> sd(nasdaq$rlatin, na.rm = TRUE)
[1] 0.03643025
```

**→ calcolo il rendimento medio giornaliero e la deviazione standard dell'indice UK**

```
> mean(nasdaq$ruk, na.rm = TRUE)
[1] 0.002245139
> sd(nasdaq$ruk, na.rm = TRUE)
[1] 0.05575466
```

**→ calcolo il rendimento medio giornaliero e la deviazione standard dell'indice US**

```
> mean(nasdaq$rus, na.rm = TRUE)
[1] 0.006999786
> sd(nasdaq$rus, na.rm = TRUE)
[1] 0.1275861
```

**b) Assumendo che il rendimento del titolo non rischioso sia nullo, calcolare gli Sharpe ratio dei vari indici. Su quale titolo conviene investire in base a questa statistica?**

Calcolo Indice di Sharpe per i 7 indici:

```
> mean(nasdaq$rasia, na.rm = TRUE)/sd(nasdaq$rasia, na.rm = TRUE)
[1] 0.03462788
> mean(nasdaq$rema, na.rm = TRUE)/sd(nasdaq$rema, na.rm = TRUE)
[1] 0.04519204
> mean(nasdaq$reuro, na.rm = TRUE)/sd(nasdaq$reuro, na.rm = TRUE)
```

```

[1] 0.04980713
> mean(nasdaq$rglobal, na.rm = TRUE)/sd(nasdaq$rglobal, na.rm = TRUE)
[1] 0.04317217
> mean(nasdaq$rlatin, na.rm = TRUE)/sd(nasdaq$rlatin, na.rm = TRUE)
[1] 0.02133111
> mean(nasdaq$ruk, na.rm = TRUE)/sd(nasdaq$ruk, na.rm = TRUE)
[1] 0.04026818
> mean(nasdaq$rus, na.rm = TRUE)/sd(nasdaq$rus, na.rm = TRUE)
[1] 0.05486325

```

**c) Ricavare media, deviazione standard e Sharpe ratio dei rendimenti di un portafoglio P avente per quote [0.1 0.1 0.1 0.3 0.1 0.1 0.2], rispettivamente negli indici Asia, EMEA, Eurozone, Global, Latin America, UK e US.**

```

> nasdaq$rPORT = (0.1*nasdaq$rasia +0.1*nasdaq$remae
+0.1*nasdaq$reuro +0.3*nasdaq$rglobal +0.1*nasdaq$rlatin
+0.1*nasdaq$ruk +0.2*nasdaq$rus)
> mean(nasdaq$rPORT, na.rm = TRUE)
[1] 0.003722605
> sd(nasdaq$rPORT, na.rm = TRUE)
[1] 0.07469904
> summary(nasdaq$rPORT, na.rm = TRUE)
      Min.      1st Qu.      Median      Mean      3rd Qu.      Max.      NA's
-0.2868984 -0.0054936  0.0003856  0.0037226  0.0070549  0.4388550      25
> mean(nasdaq$rPORT, na.rm = TRUE)/sd(nasdaq$rPORT, na.rm = TRUE)
[1] 0.0498347

```

**d) Ricavare media, deviazione standard e Sharpe ratio di un portafoglio che investe per il 60% nel portafoglio P e per la parte rimanente in un titolo non rischioso dal rendimento nullo.**

```

> rf = 0
> nasdaq$rPORT1 = (0.6*nasdaq$rPORT +0.4*rf)
> mean(nasdaq$rPORT1, na.rm = TRUE)
[1] 0.002233563
> sd(nasdaq$rPORT1, na.rm = TRUE)
[1] 0.04481943
> mean(nasdaq$rPORT1, na.rm = TRUE)/sd(nasdaq$rPORT1, na.rm = TRUE)
[1] 0.0498347

```

## ESERCIZIO 2

**Regredire i rendimenti dell'indice Global sui rendimenti di tutti gli altri indici.**

```
> reg1 = lm(rglobal ~ rasia + remea + reuro + rlatin + ruk + rus,
data=nasdaq)
```

```
> summary(reg1)
```

```
Call:
lm(formula = rglobal ~ rasia + remea + reuro + rlatin + ruk +
    rus, data = nasdaq)

Residuals:
    Min       1Q   Median       3Q      Max
-0.033367 -0.000739  0.000861  0.002755  0.024983

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0009133  0.0004758  -1.920  0.05615 .
    rasia      0.0195912  0.0278398   0.704  0.48233
    remea      0.4474119  0.1474450   3.034  0.00269 **
    reuro     -0.1916228  0.0783097  -2.447  0.01516 *
    rlatin    -0.0457743  0.0170497  -2.685  0.00779 **
    ruk        0.1489912  0.0788385   1.890  0.06005 .
    rus        0.4518489  0.0080865  55.877 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.007265 on 228 degrees of freedom
(25 osservazioni eliminate a causa di valori mancanti)
Multiple R-squared:  0.9919,    Adjusted R-squared:  0.9917
F-statistic: 4680 on 6 and 228 DF,  p-value: < 2.2e-16
```

**b) Stabilire con il supporto di un opportuno test statistico se sono uguali gli effetti dei rendimenti degli indici Eurozone e US.**

```
> lht(reg1, c("reuro=rus"), test = "Chisq")
```

```
Linear hypothesis test:
reuro - rus = 0

Model 1: restricted model
Model 2: rglobal ~ rasia + remea + reuro + rlatin + ruk + rus

    Res.Df    RSS Df Sum of Sq  Chisq Pr(>Chisq)
1     229 0.015626
2     228 0.012033  1 0.0035928 68.078 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

**c) Stabilire con il supporto di un opportuno test statistico se il modello di regressione è correttamente specificato.**

```
> resettest(reg1, power=2:3)
```

```
RESET test

data:  reg1
RESET = 18.081, df1 = 2, df2 = 226, p-value = 5.198e-08
```

**d) Stabilire con il supporto di un opportuno test statistico se il modello di regressione presenta eteroschedasticità.**

```
> white(reg1, interactions = TRUE)
```

```
# A tibble: 1 × 5
  statistic p.value parameter method alternative
  <dbl>     <dbl>     <dbl> <chr>      <chr>
1     185. 1.48e-25         27 White's Test greater
```

### ESERCIZIO 3

Condurre sei regressioni CAPM includendo la costante nella specificazione. Nelle regressioni la variabile esplicativa è il rendimento dell'indice Global, mentre la variabile dipendente è il rendimento di ciascuno degli altri indici.

```
> reg3 = lm(rasia ~ rglobal, data=nasdaq)
```

```
> summary(reg3)
```

```
Call:
lm(formula = rasia ~ rglobal, data = nasdaq)

Residuals:
    Min       1Q   Median       3Q      Max
-0.117479 -0.005945 -0.000669  0.006428  0.140183

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0002431  0.0014698  -0.165   0.869
rglobal      0.6232239  0.0183742  33.918 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02363 on 257 degrees of freedom
(1 osservazione eliminata a causa di un valore mancante)
Multiple R-squared:  0.8174,    Adjusted R-squared:  0.8167
F-statistic: 1150 on 1 and 257 DF,  p-value: < 2.2e-16
```

```
> reg4 = lm(remea ~ rglobal, data=nasdaq)
```

```
> summary(reg4)
```

```
Call:
lm(formula = remea ~ rglobal, data = nasdaq)

Residuals:
    Min       1Q   Median       3Q      Max
-0.083843 -0.006197 -0.001064  0.004617  0.123628

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.0003373  0.0015279   0.221   0.825
rglobal     0.7641606  0.0191003  40.008 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02457 on 257 degrees of freedom
(1 osservazione eliminata a causa di un valore mancante)
Multiple R-squared:  0.8617,    Adjusted R-squared:  0.8611
F-statistic: 1601 on 1 and 257 DF,  p-value: < 2.2e-16
```

```
> reg5 = lm(reuro ~ rglobal, data=nasdaq)
```

```
> summary(reg5)
```

```

Call:
lm(formula = reuro ~ rglobal, data = nasdaq)

Residuals:
    Min       1Q   Median       3Q      Max
-0.110418 -0.006908 -0.001813  0.006130  0.175006

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.0008082  0.0019988   0.404   0.686
rglobal      0.8434446  0.0249864  33.756 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03214 on 257 degrees of freedom
(1 osservazione eliminata a causa di un valore mancante)
Multiple R-squared:  0.816,    Adjusted R-squared:  0.8152
F-statistic: 1139 on 1 and 257 DF,  p-value: < 2.2e-16

```

```
> reg6 =lm(rlatin ~ rglobal, data=nasdaq)
```

```
> summary(reg6)
```

```

Call:
lm(formula = rlatin ~ rglobal, data = nasdaq)

Residuals:
    Min       1Q   Median       3Q      Max
-0.189797 -0.008766 -0.000118  0.008755  0.194100

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 5.829e-05  2.019e-03   0.029   0.977
rglobal      2.079e-01  2.524e-02  8.237 9.01e-15 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.03247 on 257 degrees of freedom
(1 osservazione eliminata a causa di un valore mancante)
Multiple R-squared:  0.2089,    Adjusted R-squared:  0.2058
F-statistic: 67.85 on 1 and 257 DF,  p-value: 9.013e-15

```

```
> reg7 =lm(ruk ~ rglobal, data=nasdaq)
```

```
> summary(reg7)
```

```

Call:
lm(formula = ruk ~ rglobal, data = nasdaq)

Residuals:
    Min       1Q   Median       3Q      Max
-0.078789 -0.005776 -0.000248  0.006373  0.086042

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 7.106e-06  1.279e-03   0.006   0.996
rglobal      6.474e-01  1.599e-02 40.492 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02056 on 257 degrees of freedom
(1 osservazione eliminata a causa di un valore mancante)
Multiple R-squared:  0.8645,    Adjusted R-squared:  0.864
F-statistic: 1640 on 1 and 257 DF,  p-value: < 2.2e-16

```

```
> reg8 =lm(rus ~ rglobal, data=nasdaq)
```

```
> summary(reg8)
```

```

Call:
lm(formula = rus ~ rglobal, data = nasdaq)

Residuals:
    Min       1Q   Median       3Q      Max
-0.078514 -0.006655 -0.001630  0.004099  0.117949

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.001845   0.001442    1.28   0.202
rglobal      1.572793   0.018066   87.06 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.02208 on 233 degrees of freedom
(25 osservazioni eliminate a causa di valori mancanti)
Multiple R-squared:  0.9702,    Adjusted R-squared:  0.97
F-statistic: 7580 on 1 and 233 DF,  p-value: < 2.2e-16

```

**c) Formulare l'ipotesi nulla alla base dei test di verifica del CAPM. Analizzando un indice alla volta, ci sono casi in cui si rifiuta l'ipotesi nulla? Cosa è possibile concludere?**

```
> lht(reg3, c("(Intercept)=0"), test = "Chisq")
```

```

Linear hypothesis test:
(Intercept) = 0

```

```

Model 1: restricted model
Model 2: rasia ~ rglobal

```

|   | Res.Df | RSS     | Df | Sum of Sq  | Chisq  | Pr(>Chisq) |
|---|--------|---------|----|------------|--------|------------|
| 1 | 258    | 0.14355 |    |            |        |            |
| 2 | 257    | 0.14354 | 1  | 1.5278e-05 | 0.0274 | 0.8686     |

```
> lht(reg4, c("(Intercept)=0"), test = "Chisq")
```

```

Linear hypothesis test:
(Intercept) = 0

```

```

Model 1: restricted model
Model 2: remea ~ rglobal

```

|   | Res.Df | RSS     | Df | Sum of Sq  | Chisq  | Pr(>Chisq) |
|---|--------|---------|----|------------|--------|------------|
| 1 | 258    | 0.15513 |    |            |        |            |
| 2 | 257    | 0.15511 | 1  | 2.9418e-05 | 0.0487 | 0.8253     |

```
> lht(reg5, c("(Intercept)=0"), test = "Chisq")
```

```

Linear hypothesis test:
(Intercept) = 0

```

```

Model 1: restricted model
Model 2: reuro ~ rglobal

```

|   | Res.Df | RSS     | Df | Sum of Sq  | Chisq  | Pr(>Chisq) |
|---|--------|---------|----|------------|--------|------------|
| 1 | 258    | 0.26560 |    |            |        |            |
| 2 | 257    | 0.26543 | 1  | 0.00016886 | 0.1635 | 0.686      |

```
> lht(reg6, c("(Intercept)=0"), test = "Chisq")
```

```
Linear hypothesis test:
(Intercept) = 0
```

```
Model 1: restricted model
Model 2: rlatin ~ rglobal
```

```
Res.Df    RSS Df Sum of Sq Chisq Pr(>Chisq)
1      258 0.27089
2      257 0.27089  1 8.7846e-07 8e-04      0.977
```

```
> lht(reg7, c("(Intercept)=0"), test = "Chisq")
```

```
Linear hypothesis test:
(Intercept) = 0
```

```
Model 1: restricted model
Model 2: ruk ~ rglobal
```

```
Res.Df    RSS Df Sum of Sq Chisq Pr(>Chisq)
1      258 0.10868
2      257 0.10868  1 1.3053e-08      0      0.9956
```

```
> lht(reg8, c("(Intercept)=0"), test = "Chisq")
```

```
Linear hypothesis test:
(Intercept) = 0
```

```
Model 1: restricted model
Model 2: rus ~ rglobal
```

```
Res.Df    RSS Df Sum of Sq Chisq Pr(>Chisq)
1      234 0.1144
2      233 0.1136  1 0.00079863 1.638      0.2006
```

#### d) Determinare i rendimenti storici e di equilibrio dei sei indici.

Determino i rendimenti storici:

```
> nasdaq =select(nasdaq, rasia, remea, reuro, rlatin, ruk, rus)
```

```
> summary(nasdaq)
```

| rasia    |             | remaa    |             | reuro    |            |
|----------|-------------|----------|-------------|----------|------------|
| Min.     | :-2.581e-01 | Min.     | :-0.2722032 | Min.     | :-0.308117 |
| 1st Qu.: | -7.873e-03  | 1st Qu.: | -0.0070275  | 1st Qu.: | -0.008746  |
| Median : | 6.440e-06   | Median : | -0.0001808  | Median : | -0.001002  |
| Mean :   | 1.911e-03   | Mean :   | 0.0029791   | Mean :   | 0.003724   |
| 3rd Qu.: | 9.200e-03   | 3rd Qu.: | 0.0083554   | 3rd Qu.: | 0.009519   |
| Max. :   | 3.247e-01   | Max. :   | 0.4013111   | Max. :   | 0.459923   |
| NA's :   | 1           | NA's :   | 1           | NA's :   | 1          |

  

| rlatin   |             | ruk      |             | rus      |             |
|----------|-------------|----------|-------------|----------|-------------|
| Min.     | :-0.2386743 | Min.     | :-0.2279542 | Min.     | :-0.4387771 |
| 1st Qu.: | -0.0088131  | 1st Qu.: | -0.0079205  | 1st Qu.: | -0.0083345  |
| Median : | -0.0001574  | Median : | 0.0004304   | Median : | 0.0006212   |
| Mean :   | 0.0007771   | Mean :   | 0.0022451   | Mean :   | 0.0069998   |
| 3rd Qu.: | 0.0104287   | 3rd Qu.: | 0.0075392   | 3rd Qu.: | 0.0125125   |
| Max. :   | 0.2245207   | Max. :   | 0.3342201   | Max. :   | 0.8612135   |
| NA's :   | 1           | NA's :   | 1           | NA's :   | 25          |

## ESERCIZIO 4

**a) Condurre un test Ljung-Box per verificare la presenza di autocorrelazione fino all'ordine 5. Cosa si può concludere?**

```
> Box.test(nasdaq$rglobal, lag = 5, type = "Ljung-Box")
```

Box-Ljung test

```
data: nasdaq$rglobal
X-squared = 74.201, df = 5, p-value = 1.366e-14
```

**b) Determinare in base al criterio AIC il migliore tra i modelli AR, MA e ARMA fino al modello ARMA(2,2).**

```
> rglobal_clean = na.omit(nasdaq$rglobal) → per rimuovere i valori NA
```

```
> ar1 = arma(rglobal_clean, order=c(1,0))
```

```
> summary(ar1)
```

```
Call:
arma(x = rglobal_clean, order = c(1, 0))

Model:
ARMA(1,0)

Residuals:
    Min       1Q   Median       3Q      Max
-0.297002 -0.009230 -0.002946  0.003461  0.467826

Coefficient(s):
            Estimate Std. Error t value Pr(>|t|)
ar1      -0.004555    0.062135  -0.073   0.942
intercept  0.003506    0.004980   0.704   0.481

Fit:
sigma^2 estimated as 0.006436, Conditional Sum-of-Squares = 1.65, AIC = -567.5
```

```
> ar2 = arma(rglobal_clean, order=c(2,0))
```

```
> summary(ar2)
```

```
Call:
arma(x = rglobal_clean, order = c(2, 0))

Model:
ARMA(2,0)

Residuals:
    Min       1Q   Median       3Q      Max
-0.296920 -0.009119 -0.002822  0.002936  0.467835

Coefficient(s):
            Estimate Std. Error t value Pr(>|t|)
ar1      -0.004454    0.062130  -0.072   0.943
ar2       0.010712    0.062128   0.172   0.863
intercept  0.003416    0.004994   0.684   0.494

Fit:
sigma^2 estimated as 0.00646, Conditional Sum-of-Squares = 1.65, AIC = -564.9
```

```
> ma1 = arma(rglobal_clean, order=c(0,1))
```

```
> summary(ma1)
```

```
Call:
arma(x = rglobal_clean, order = c(0, 1))

Model:
ARMA(0,1)

Residuals:
      Min       1Q   Median       3Q      Max
-0.297014 -0.009225 -0.002959  0.003452  0.467816

Coefficient(s):
      Estimate Std. Error t value Pr(>|t|)
mal      -0.004439   0.061472  -0.072   0.942
intercept  0.003502   0.004953   0.707   0.480

Fit:
sigma^2 estimated as 0.006436, Conditional Sum-of-Squares = 1.65, AIC = -567.9
```

```
> ma2 =arma(rglobal_clean, order=c(0,2))
```

```
> summary(ma2)
```

```
Call:
arma(x = rglobal_clean, order = c(0, 2))

Model:
ARMA(0,2)

Residuals:
      Min       1Q   Median       3Q      Max
-0.296919 -0.009088 -0.002820  0.003025  0.467825

Coefficient(s):
      Estimate Std. Error t value Pr(>|t|)
mal      -0.004919   0.062184  -0.079   0.937
ma2       0.011294   0.063044   0.179   0.858
intercept  0.003435   0.005016   0.685   0.493

Fit:
sigma^2 estimated as 0.00646, Conditional Sum-of-Squares = 1.65, AIC = -564.91
```

```
> arma11 =arma(rglobal_clean, order=c(1,1))
```

```
> summary(arma11)
```

```
Call:
arma(x = rglobal_clean, order = c(1, 1))

Model:
ARMA(1,1)

Residuals:
      Min       1Q   Median       3Q      Max
-0.303607 -0.014613 -0.002446  0.009077  0.461674

Coefficient(s):
      Estimate Std. Error t value Pr(>|t|)
ar1      -0.923816   0.065091 -14.193 <2e-16 ***
ma1       0.872345   0.079069  11.033 <2e-16 ***
intercept  0.005989   0.009169   0.653   0.514
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fit:
sigma^2 estimated as 0.006333, Conditional Sum-of-Squares = 1.63, AIC = -570.9
```

```
> arma12 =arma(rglobal_clean, order=c(1,2))
```

```
> summary(arma12)
```

```
Call:
arma(x = rgloba_clean, order = c(1, 2))

Model:
ARMA(1,2)

Residuals:
    Min       1Q   Median       3Q      Max
-0.296892 -0.009122 -0.002784  0.002964  0.467818

Coefficient(s):
            Estimate Std. Error t value Pr(>|t|)
ar1         0.146005   0.847320   0.172   0.863
ma1        -0.150817   0.845738  -0.178   0.858
ma2         0.013863   0.076763   0.181   0.857
intercept   0.002917   0.005269   0.554   0.580

Fit:
sigma^2 estimated as 0.006459, Conditional Sum-of-Squares = 1.65, AIC = -562.5
```

```
> arma21 = arma(rgloba_clean, order=c(2,1))
```

```
> summary(arma21)
```

```
Call:
arma(x = rgloba_clean, order = c(2, 1))

Model:
ARMA(2,1)

Residuals:
    Min       1Q   Median       3Q      Max
-0.296892 -0.009073 -0.002783  0.002947  0.467868

Coefficient(s):
            Estimate Std. Error t value Pr(>|t|)
ar1         0.146717   1.344719   0.109   0.913
ar2         0.009631   0.076000   0.127   0.899
ma1        -0.151201   1.343561  -0.113   0.910
intercept   0.002874   0.006304   0.456   0.648

Fit:
sigma^2 estimated as 0.00646, Conditional Sum-of-Squares = 1.65, AIC = -562.92
```

```
> arma22 = arma(rgloba_clean, order=c(2,2))
```

```
> summary(arma22)
```

```
Model:
ARMA(2,2)

Residuals:
    Min       1Q   Median       3Q      Max
-0.272664 -0.024501 -0.002301  0.015523  0.408551

Coefficient(s):
            Estimate Std. Error t value Pr(>|t|)
ar1         0.519751      NaN      NaN      NaN
ar2        -0.561822   0.051001  -11.016 <2e-16 ***
ma1        -0.620526      NaN      NaN      NaN
ma2         0.860991      NaN      NaN      NaN
intercept   0.004248   0.005792   0.733   0.463
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Fit:
sigma^2 estimated as 0.005787, Conditional Sum-of-Squares = 1.48, AIC = -589.5
```

**c) Condurre un test di radice unitaria (augmented Dickey-Fuller) con costante e trend. Cosa si può concludere?**

```
> adf.test(nasdaq$rglobal)
```

```
Type 3: with drift and trend
lag    ADF p.value
[1,]    0 -16.06    0.01
[2,]    1 -11.17    0.01
[3,]    2  -9.09    0.01
[4,]    3  -7.91    0.01
[5,]    4 -13.00    0.01
```